Cryptography and Network Security (Spring 2026) – Term Paper

This assessment is worth **15%** of your course credits.

Post Date: 25th March 2025 Submission Due: 30th April 2026 (Midnight. Online via LMS)

Number of weeks to complete this assessment: 05 (week 11 – week 16)

Term Paper Viva: Wk16.

No Submission will be allowed after Thursday 30th April.

In this assessment, you are required to write a term paper on a topic relevant to Cryptography and Network Security course. For this research, you can make a group of up to 4 students. (5 students can be allowed as an exception.)

About the Term Paper:

A term paper is ideally a well-researched, well-written academic text that features literature review, critical analysis and interpretation of a selected subject matter or topic.

Knowing how to write a term paper is in itself an important qualification and an essential part of your academic career. In a term paper you can show:

- that you can formulate a research question,
- and that you are able to analyze and compare different techniques, algorithms and methods
- that you know how to research a subject and that you have read as well as understood academic publications on your subject matter
- that you have a firm grasp of the English language and that you can employ an academic writing style.

The type of term paper you will write in the CNS course will closely resemble a literature review or survey paper related to a technical topic. Those students who are interested in developing new techniques can also propose and include new information and methods in their paper.

Note:
Do not underestimate the time it takes to research and select literature, develop your argument and thesis, proofread your paper, etc. A term paper cannot be written in three days! Researching and surveying literature alone will take a considerable amount of time. While writing your term paper, you may need to go back to researching literature if what you have found earlier is not helpful for your topic. You may have to reconsider your general take on your subject or revise your argument.

Points to Remember regarding submission:

1. Use the Overleaf/IEEE conference paper template for the paper write-up, and share the link with the class instructor when submitting the paper through LMS, and also submit a PDF of the final version via LMS. The IEEE templates in Word or Tex formats are available from the IEEE link. ieee.org/conferences/publishing/templates.html
2. Try to limit your write-up to 10 pages at maximum, back and front pages of a single paper are counted as two pages.
3. The references need to be added in IEEE referencing style. Refer to (https://www.cite.auckland.ac.nz/2_9.html).

4. It is highly advised to include recent papers (past 1-3 years). Keep your paper well organized and add proper captions for all figures and tables.
5. The submission will be online and it will be checked through Turnitin for similarity check. The similarity percentage must be less than 20% (overall) and less than 5% with one source. Any write-up not meeting these rules will be penalized in grading, while similarity greater than 40% will not be graded at all.

Topics from CNS Knowledge Domain:

Some topics are as follows, but you are not restricted to selecting your topic only from this list:

- Elliptic Curve Cryptography and Digital Signatures
- IoT Security and Lightweight Cryptography (Ascon)
- Quantum Cryptography
- Security for Industrial Control Systems / SCADA networks security
- Machine Learning in Firewall Development
- Watermarking and Steganography
- Incident Response System
- Intrusion Detection and Prevention Schemes
- Zero trust security model
- Cryptanalysis of Symmetric Key Cryptosystems
- Security technologies in Cloud Computing
- Enterprise VPNs using GRE, MPLS, IPSec
- Endpoint security technologies
- Security technologies used in Smartphones
- Secure File Systems
- Reconfigurable Crypto-processors
- Post-Quantum Cryptography
- Stream Cipher Types and Techniques
- Security at the Transport Layer (TLS and SSL)
- Security at the Application Layer (PGP and S/MIME)
- Hashing techniques used in Blockchain
- Security in IoT or Lightweight Cryptography (ASCON)

Structure of a Term Paper:

I have discussed this structure of term paper in the past with many individual groups. This is presented here just as a guide. Following sections can be used in your term paper:

1. Abstract

2. Introduction

   The last paragraph of introduction should include the following sentence. "The rest of the paper is organized in such a way that Section 3 discusses related work found in literature, followed by a discussion in Section 4 and the paper is concluded in Section 5."

3. Related Work / Literature Review

   3.1 Classification1 / Grouping1

   3.1.1. Technique 1/ Method 1

   3.1.2 Technique 2/ Method 2

   3.1.3 Technique 3 / Method 3

   3.2 Classification 2 / Grouping 2

   3.2.1 Technique 1/ Method 1

   3.2.2 Technique 2/ Method 2

   3.2.3 Technique 3 / Method 3

4. Discussion (Comparison of Strengths and Weaknesses of above Techniques)

5. Conclusion / Summary

6. References

7. Bibliography / Additional Reading References

Rubric used for grading:

Rubric for Term Paper				
Criteria	**Exemplary**	**Good**	**Acceptable**	**Unacceptable**
Purpose	The writer's central purpose or argument is readily apparent to the reader.	The writing has a clear purpose or argument, but may sometimes digress from it.	The central purpose or argument is not consistently clear throughout the paper.	The purpose or argument is generally unclear.
Content	Balanced presentation of relevant and legitimate information that clearly supports a central purpose or argument and shows a thoughtful, in-depth analysis of a significant topic. Reader gains important insights.	Information provides reasonable support for a central purpose or argument and displays evidence of a basic analysis of a significant topic. Reader gains some insights.	Information supports a central purpose or argument at times. Analysis is basic or general. Reader gains few insights.	Central purpose or argument is not clearly identified. Analysis is vague or not evident. Reader is confused or may be misinformed.
Organization	The ideas are arranged logically to support the purpose or argument. They flow smoothly from one to another and are clearly linked to each other. The reader can follow the line of reasoning.	The ideas are arranged logically to support the central purpose or argument. They are usually clearly linked to each other. For the most part, the reader can follow the line of reasoning.	In general, the writing is arranged logically, although occasionally ideas fail to make sense together. The reader is fairly clear about what writer intends.	The writing is not logically organized. Frequently, ideas fail to make sense together. The reader cannot identify a line of reasoning and loses interest.
Feel	The writing is compelling. It hooks the reader and sustains interest throughout.	The writing is generally engaging, but has some dry spots. In general, it is focused and keeps the reader's attention.	The writing is dull and unengaging. Though the paper has some interesting parts, the reader finds it difficult to maintain interest.	The writing has little personality. The reader quickly loses interest and stops reading.

Criteria	Exemplary	Good	Acceptable	Unacceptable
Tone	The tone is consistently professional and appropriate for an academic research paper.	The tone is generally professional. For the most part, it is appropriate for an academic research paper.	The tone is not consistent-ly professional or appropriate for an academic research paper.	The tone is unprofessional. It is not appropriate for an academic research paper.
Sentence Structure	Sentences are well-phrased and varied in length and structure. They flow smoothly from one to another.	Sentences are well-phrased and there is some variety in length and structure. The flow from sentence to sentence is generally smooth.	Some sentences are awkwardly constructed so that the reader is occasionally distracted.	Errors in sentence structure are frequent enough to be a major distraction to the reader.
Word Choice	Word choice is consistently precise and accurate.	Word choice is generally good. The writer often goes beyond the generic word to find one more precise and effective.	Word choice is merely adequate, and the range of words is limited. Some words are used inappropriately.	Many words are used inappropriately, confusing the reader.
Grammar, Spelling, Writing Mechanics (punctuation, italics, capitalization etc.	The writing is free or almost free of errors.	There are occasional errors, but they don't represent a major distraction or obscure meaning.	The writing has many errors, and the reader is distracted by them.	There are so many errors that meaning is obscured. The reader is confused and stops reading.

Criteria	Exemplary	Good	Acceptable	Unacceptable
Use of References	Compelling evidence from professionally legitimate sources is given to support claims. Attribution is clear and fairly represented.	Professionally legitimate sources that support claims are generally present and attribution is, for the most part, clear and fairly represented.	Although attributions are occasionally given, many statements seem unsubstantiated. The reader is confused about the source of information and ideas.	References are seldom cited to support statements.
Quality of References	References are primarily peer- reviewed professional journals or other approved sources (e.g., government documents, agency manuals, …). The reader is confident that the information and ideas can be trusted.	Although most of the references are professionally legitimate, a few are questionable (e.g., trade books, internet sources, popular magazines, …). The reader is uncertain of the reliability of some of the sources.	Most of the references are from sources that are not peer-reviewed and have uncertain reliability. The reader doubts the accuracy of much of the material presented.	There are virtually no sources that are professionally reliable. The reader seriously doubts the value of the material and stops reading.
Use of IEEE format for referencing	IEEE format is used accurately and consistently in the paper and in the "References" section.	IEEE format is used with minor errors.	There are frequent errors in IEEE format.	Format of the document is not recognizable as IEEE.